

6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

```
clc; clear; close all
```

```
n=0:79;
```

```
x=sin(2*pi*n/40);
```

```
q=8;
```

```
xt=floor(x*q)/q;
```

```
xr=round(x*q)/q;
```

```
et=x-xt;
```

```
er=x-xr;
```

```
% Fig 1
```

```
figure
```

```
plot(n,x,'b',n,xr,'r','LineWidth',1.5)
```

```
grid on
```

```
title('Original vs Quantized')
```

```
legend('Original','Quantized')
```

```
% Fig 2
```

```
figure
```

```
plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)
```

```
grid on
```

```
title('Truncation vs Rounding')
```

```
legend('Original','Truncated','Rounded')
```

```
% Fig 3
```

```
figure
```

```
stem(n,et,'r')
```

```
hold on
```

```
stem(n,er,'g')
```

```
grid on
```

```
title('Quantization Error')
```

```
legend('Truncation','Rounding')
```

```
% Fig 4
```

figure

```
bar([mean(et.^2) mean(er.^2)])
```

```
set(gca,'XTickLabel',{'Truncation','Rounding'})
```

grid on

```
title('MSE Comparison')
```

```
ylabel('MSE')
```





